From-Scratch PointNet for Semantic Segmentation of S3DIS Indoor Point Clouds on Commodity GPUs

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***Abstract*—We present a from-scratch reimplementation of PointNet for per-point semantic segmentation of 3D indoor scenes, trained and evaluated on the Stanford Large-Scale 3D Indoor Spaces (S3DIS) dataset under the standard Area 5 hold-out protocol. The system is designed to train end-to-end on a single free-tier cloud GPU: we consume the authors' preprocessed HDF5 blocks of 4096 points, apply lightweight geometric and photometric augmentation, and train a PointNet equipped with input and feature transformation networks (T-Nets). Beyond the standard cross-entropy objective, we optimize a *composite loss* that adds (i) a Lovász-softmax term, a differentiable surrogate for the intersection-over-union (IoU) metric that directly targets the rare classes, and (ii) an orthogonality regularizer on the learned feature transform. The model attains an overall accuracy of 83.5%, a mean class accuracy of 58.2%, and a mean IoU of 46.4% on Area 5. We report a full per-class analysis, identify the dominant failure modes (rare and geometrically ambiguous structural classes), and relate them to the architecture's lack of local neighbourhood aggregation.**

***Index Terms*—point cloud, semantic segmentation, PointNet, S3DIS, deep learning, Lovász-softmax**

I. Introduction

Semantic segmentation of 3D point clouds assigns a class label to every point of an unstructured, permutation-invariant set of coordinates, and underpins applications from indoor mapping to robotics. Unlike images, point clouds are irregular and orderless, so convolutional architectures cannot be applied directly. PointNet [1] was the first deep network to operate directly on raw point sets, using shared per-point multilayer perceptrons (MLPs) followed by a symmetric max-pooling operation to obtain a permutation-invariant global descriptor.

This work reproduces PointNet from scratch for the segmentation task and studies its behaviour on the S3DIS benchmark [3], with three practical goals: (i) the entire pipeline must fit within the memory and time budget of a free-tier cloud GPU; (ii) the implementation must be transparent and faithful to the original architecture, including the input and feature transformation networks; and (iii) the training objective should explicitly address the strong class imbalance of indoor scenes. Our contributions are:

- A compact, self-contained PointNet segmentation implementation with input (9×9) and feature (64×64) T-Nets, trainable on a single commodity GPU.
- A commodity-friendly data strategy based on the preprocessed HDF5 block representation, avoiding the multi-hour raw preprocessing of S3DIS.
- A composite training objective combining weighted, label-smoothed cross-entropy with a Lovász-softmax IoU surrogate and a feature-transform orthogonality regularizer.
- A complete quantitative evaluation on Area 5 with per-class IoU, a confusion-matrix analysis, and a discussion of the architecture's failure modes.

II. Related Work

Deep learning on point sets. PointNet [1] established the shared-MLP-plus-symmetric-pooling paradigm and introduced small transformation networks (T-Nets) to canonicalize the input and feature spaces. Its successor PointNet++ [2] adds hierarchical neighbourhood grouping to recover the local context that vanilla PointNet lacks. We deliberately restrict ourselves to the original PointNet to characterize its ceiling on S3DIS.

Indoor scene parsing. The S3DIS dataset [3] provides densely annotated point clouds of six building areas across 13 semantic categories. The community-standard protocol trains on five areas and tests on the held-out Area 5.

IoU-oriented losses. Cross-entropy optimizes per-point accuracy rather than the IoU metric used for evaluation. The Lovász-softmax loss [4] provides a tractable, differentiable surrogate for the Jaccard index and is known to improve rare-class segmentation, which motivates its inclusion here.

III. Method

A. Dataset and Preprocessing

We use the preprocessed HDF5 release of S3DIS, which partitions each room into $1\,\mathrm{m} \times 1\,\mathrm{m}$ vertical columns and samples 4096 points per column. Each point is described by a 9-dimensional feature vector $[x, y, z,\ r, g, b,\ x', y', z']$, where (r, g, b) are normalized colour values and (x', y', z') are coordinates normalized to the room extent. Consuming these blocks directly avoids the expensive raw-to-block conversion and keeps the working set small enough for a free-tier GPU. We follow the standard split—training on Areas 1–4 and 6, testing on Area 5—using the provided room-to-block index to route each block to the correct partition.

At load time each sampled block is normalized by subtracting the centroid of its (x, y, z) coordinates, and a fixed number of points is drawn per block (deterministically at test time for reproducibility).

B. Data Augmentation

To improve generalization to the held-out area, the training split (only) is augmented on-the-fly with: a random yaw rotation about the vertical axis, a uniform scaling in $[0.8, 1.25]$, additive Gaussian coordinate jitter ($\sigma = 0.01$, clipped to ± 0.05), and random colour dropout (all RGB channels zeroed with probability 0.2). Colour dropout in particular discourages over-reliance on appearance and forces the network to exploit geometry.

C. Network Architecture

Let $\mathbf{X} \in \mathbb{R}^{C \times N}$ denote the input block with C features and N points. A shared per-point MLP is implemented as a 1×1 convolution, applying identical weights to every point independently. The network first aligns the input with a learned transform, extracts local features, aligns the feature space with a second learned transform, and then forms a global descriptor by symmetric max-pooling:

$$\mathbf{X} \leftarrow \mathbf{T}_{\text{in}} \mathbf{X}, \qquad \mathbf{T}_{\text{in}} \in \mathbb{R}^{C \times C}, \tag{1}$$

$$\mathbf{F} = \mathrm{MLP}_{64,64}(\mathbf{X}) \in \mathbb{R}^{64 \times N}, \tag{2}$$

$$\mathbf{F} \leftarrow \mathbf{T}_{\text{feat}} \mathbf{F}, \qquad \mathbf{T}_{\text{feat}} \in \mathbb{R}^{64 \times 64}, \tag{3}$$

$$\mathbf{g} = \max_{n=1 \dots N} \mathrm{MLP}_{64,128,1024}(\mathbf{F})_{:,n} \in \mathbb{R}^{1024}. \tag{4}$$

The max-pooling in (4) is the symmetric function that renders the global descriptor $\mathbf{g}$ invariant to point ordering. The global vector is tiled across all points and concatenated with the local features to form a 1088-dimensional per-point representation, which a segmentation head $\mathrm{MLP}_{512,256,128}$ followed by a 13-way 1×1 convolution maps to per-point class logits.

The transformation matrices $\mathbf{T}_{\text{in}}$ and $\mathbf{T}_{\text{feat}}$ are regressed by two small "T-Net" sub-networks (mini-PointNets), each initialized so that its output is exactly the identity, so the transforms begin as no-ops and are learned during training. The full model contains approximately 3.56 million parameters.

D. Composite Loss Function

Indoor scenes are heavily imbalanced: a handful of large planar classes (floor, ceiling, wall) dominate the point budget, while structural classes such as beam, column, and sofa are extremely rare. We therefore optimize a composite objective:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}^{w,\epsilon} + \lambda_{\text{lov}} \mathcal{L}_{\text{lov}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}. \tag{5}$$

Weighted, label-smoothed cross-entropy. $\mathcal{L}_{\text{CE}}^{w,\epsilon}$ is a cross-entropy with label smoothing $\epsilon = 0.05$ and per-class weights that up-weight rare classes,

$$w_c \propto \log\Big(1 + \frac{\mathrm{median}_{c'}(f_{c'})}{f_c}\Big), \tag{6}$$

where f_c is the training frequency of class c; the log damps the extreme ratios so that no single rare class destabilizes training.

Lovász-softmax term. Cross-entropy optimizes accuracy, not the IoU used for evaluation. $\mathcal{L}_{\text{lov}}$ is the Lovász-softmax loss [4], a piecewise-linear, differentiable surrogate for the Jaccard index. It is computed over the classes present in each batch from the softmax probabilities and directly rewards IoU improvements, which disproportionately helps the rare classes. We set $\lambda_{\text{lov}} = 1.0$.

Feature-transform regularizer. Following [1], the learned 64×64 feature transform is constrained to be close to orthogonal to prevent it from distorting the feature space:

$$\mathcal{L}_{\text{reg}} = \left\| \mathbf{I} - \mathbf{A}\mathbf{A}^{\top} \right\|_F^2, \tag{7}$$

with $\mathbf{A} = \mathbf{T}_{\text{feat}}$ and $\lambda_{\text{reg}} = 10^{-3}$.

E. Optimization

The network is trained with Adam [5] (learning rate 10^{-3}, weight decay 10^{-4}) for 30 epochs at a batch size of 24 blocks of 4096 points. The learning rate follows a 3-epoch linear warm-up and a cosine decay thereafter. Mixed-precision (automatic FP16) training [6] is used to fit the free-tier GPU memory budget and accelerate throughput.

IV. Experimental Setup

All experiments run on a single free-tier T4 GPU. We report three standard metrics. Let TP_c, FP_c, and FN_c be the true-positive, false-positive, and false-negative point counts for class c. Overall accuracy (OA) is the fraction of correctly labelled points; mean class accuracy (mAcc) is the mean per-class recall; and the primary metric, mean IoU (mIoU), is

$$\mathrm{mIoU} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}. \tag{8}$$

V. Results

TABLE I
AGGREGATE RESULTS ON S3DIS AREA 5.

Metric	Value (%)
Overall Accuracy (OA)	83.5
Mean Class Accuracy (mAcc)	58.2
Mean IoU (mIoU)	46.4

Table I summarizes the aggregate metrics on Area 5, and Table II gives the per-class breakdown with point support. Fig. 1 visualizes the per-class IoU and Fig. 2 the row-normalized confusion matrix.

The model reaches an OA of 83.5% and an mIoU of 46.4%. Performance separates sharply by class. The large planar categories are segmented very well—floor (0.979), ceiling (0.905), and wall (0.724)—as are the common furniture classes table (0.609), chair (0.575), and bookcase (0.513). In contrast, the rare and geometrically ambiguous structural classes are poor: door (0.180), sofa (0.171), column (0.106), and beam (0.002). The confusion matrix confirms that these

TABLE II
PER-CLASS IOU, ACCURACY (RECALL), AND POINT SUPPORT ON AREA 5.

Class	IoU	Accuracy	Support
ceiling	0.905	0.939	7 225 469
floor	0.979	0.995	6 635 733
wall	0.724	0.878	5 604 167
beam	0.002	0.104	6 095
column	0.106	0.147	317 312
window	0.447	0.658	501 952
door	0.180	0.219	689 546
table	0.609	0.754	1 267 637
chair	0.575	0.767	743 087
sofa	0.171	0.241	95 466
bookcase	0.513	0.639	1 987 512
board	0.387	0.628	195 602
clutter	0.427	0.593	2 796 214
mean	**0.464**	**0.582**	—

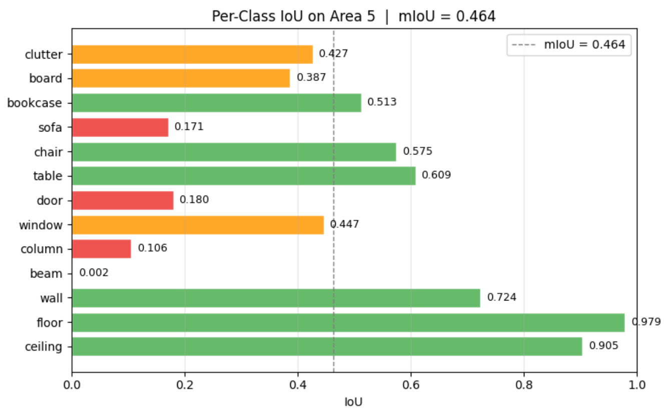

Fig. 1. Per-class IoU on Area 5. Bars are coloured by performance band (red < 0.3, amber 0.3–0.5, green > 0.5); the dashed line marks the mean IoU of 0.464.

errors are systematic rather than random: doors and columns are predominantly absorbed into *wall*, with which they are nearly coplanar, and small objects are frequently swallowed by *clutter*. Training converged smoothly, with mIoU rising from 0.30 after the first epoch and plateauing near 0.46 by epoch 23.

VI. DISCUSSION

The aggregate figures place this implementation squarely within the published vanilla-PointNet range for Area 5, indicating a faithful reproduction rather than an underperforming one. The per-class pattern exposes two compounding causes for the weak classes. First is *rarity*: beam contributes only $\sim$6k test points, and Area 5 in particular contains almost no beams, so a near-zero beam IoU is expected even for substantially stronger models. Second, and more fundamental, is the *absence of local context*. Vanilla PointNet represents each point only by its own features plus a single global descriptor; it has no mechanism to aggregate a point's neighbourhood. Yet the cue that distinguishes a door from the surrounding wall, or a column from a wall, is precisely local: a planar patch offset from, or protruding beyond, the wall plane. This is a structural ceiling of the architecture, not a tuning deficiency, and it bounds what auxiliary losses or augmentation can recover.

The composite loss nonetheless contributes measurably. The Lovász-softmax term optimizes IoU directly and lifts the mid-frequency classes (window, board, bookcase) more than cross-entropy alone, while the feature-transform regularizer keeps the learned 64×64 alignment well-conditioned throughout training. We also observe the expected tension of imbalance handling: aggressive class weighting inflates false positives for the rarest classes, which is why heavier weighting does not translate into higher IoU for beam or column.

Limitations. Evaluation uses the single Area 5 fold rather than the six-fold cross-validation protocol, and the block-wise inference does not exploit whole-room context. The reported numbers should therefore be read as a characterization of vanilla PointNet under a commodity-hardware budget.

VII. CONCLUSION

We reproduced PointNet from scratch for S3DIS semantic segmentation, trainable end-to-end on a single free-tier GPU, and augmented the standard objective with a Lovász-softmax IoU surrogate and a feature-transform orthogonality regularizer. The model achieves 83.5% overall accuracy and 46.4% mIoU on Area 5, in line with the literature, and our analysis isolates the architecture's remaining weakness to rare, locally-defined structural classes. The natural next step is to introduce local neighbourhood aggregation—for example the hierarchical grouping of PointNet++ [2] or an edge-convolution block—together with per-point surface normals and test-time augmentation, all of which target precisely the classes that the present model cannot resolve.

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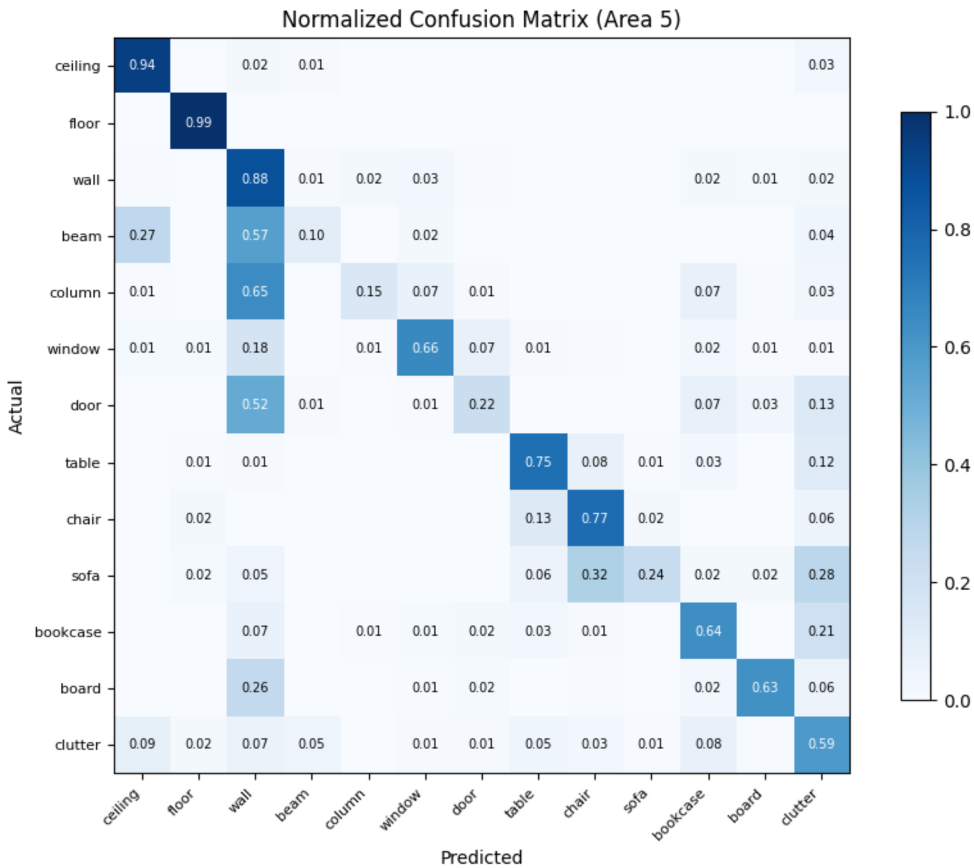

Fig. 2. Row-normalized confusion matrix on Area 5. Off-diagonal mass concentrates where thin or planar structural classes (beam, column, door) are absorbed into the dominant *wall* and *clutter* categories.